PYTHON MINI PROJECT

STUDENT PERFORMANCE ANALYTICS SYSTEM

PROJECT DOCUMENTATION

A Python-based application for analysing student marks and attendance using Pandas and NumPy.

SUBMITTED BY

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Technologies: Python • Pandas • NumPy • CSV

1. Project Overview

The Student Performance Analytics System reads student records from a CSV dataset and analyses subject marks and attendance. It calculates totals and averages, assigns grades, identifies pass/fail status, summarises overall and subject-wise performance, and exports the processed results.

2. Objectives

• Load student data from a CSV file using Pandas.

• Calculate total marks and average marks for each student.

• Assign grades and determine pass/fail status using the program's conditions.

• Summarise class performance and subject-wise averages.

• Identify the top three performers.

• Use NumPy to calculate overall mean, maximum, and minimum marks.

• Save analysed records to **student_results.csv**.

3. Technologies Used

Technology	Purpose
Python	Program logic, functions, loops, conditions, and formatted output.
Pandas	Reading and processing CSV data in a DataFrame.
NumPy	Numerical summary calculations.
CSV	Input dataset and exported analysis results.

4. Dataset Description

The dataset contains 20 student records. Fields shown in the output include Student_ID, Name, Department, Machine_Learning, Java, Mathematics, Operating_Systems, Artificial_Intelligence, and Attendance.

5. Main Features

• Dataset preview and individual student performance summary.

• Total, average, grade, attendance, and pass/fail result for each student.

• Class average, highest and lowest average, and pass/fail counts.

• Subject-wise averages and best/lowest subject average.

• Top three performers and NumPy summary statistics.

• Export of the processed analysis to student_results.csv.

6. Implementation Summary

The program loads the student CSV file, processes subject marks, calculates total and average marks, applies grade and result conditions, produces class and subject summaries, sorts students to find the top three performers, and writes the processed table to student_results.csv.

Result logic shown by the output

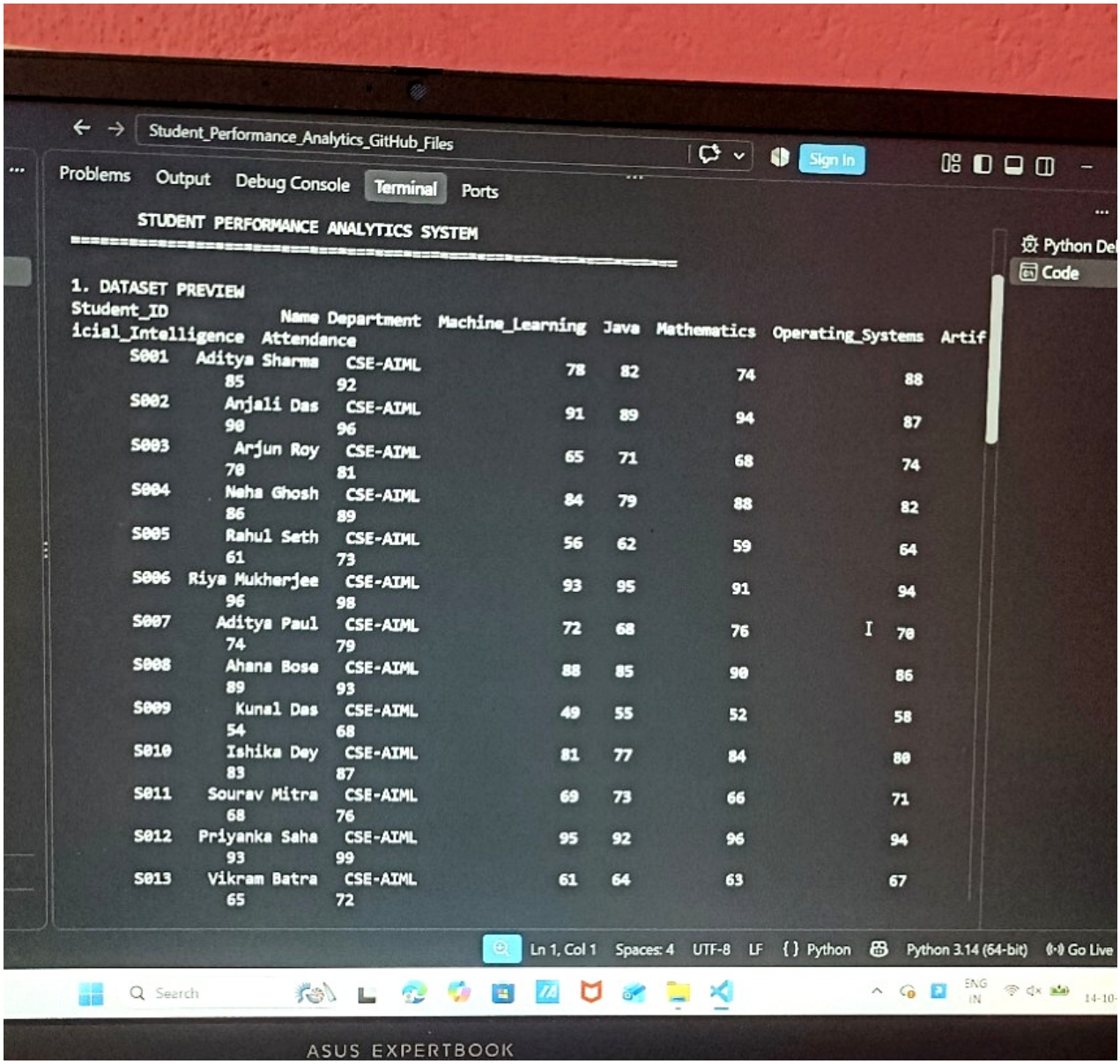
The displayed output marks a student as Pass when the average is at least 50 and attendance is at least 75; otherwise, the result is Fail. Grade labels are assigned according to the thresholds coded in the project.

7. Program Output Screenshots

The following pages present the supplied VS Code terminal output in a consistent, labelled layout. The images have been cropped to focus on the program output.

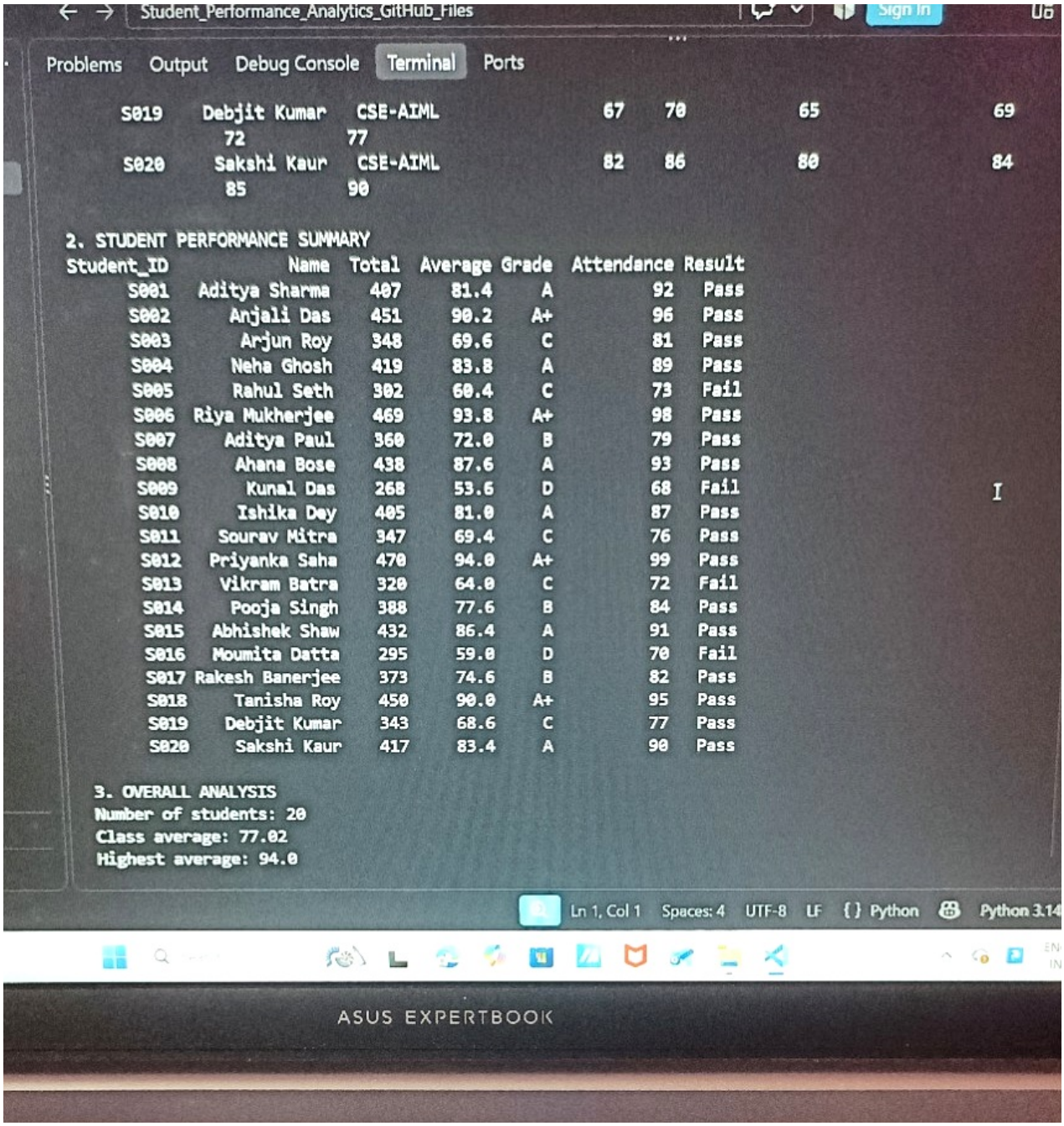
Screenshot 1 — Dataset Preview

Shows the student dataset, subject-wise marks, and attendance.



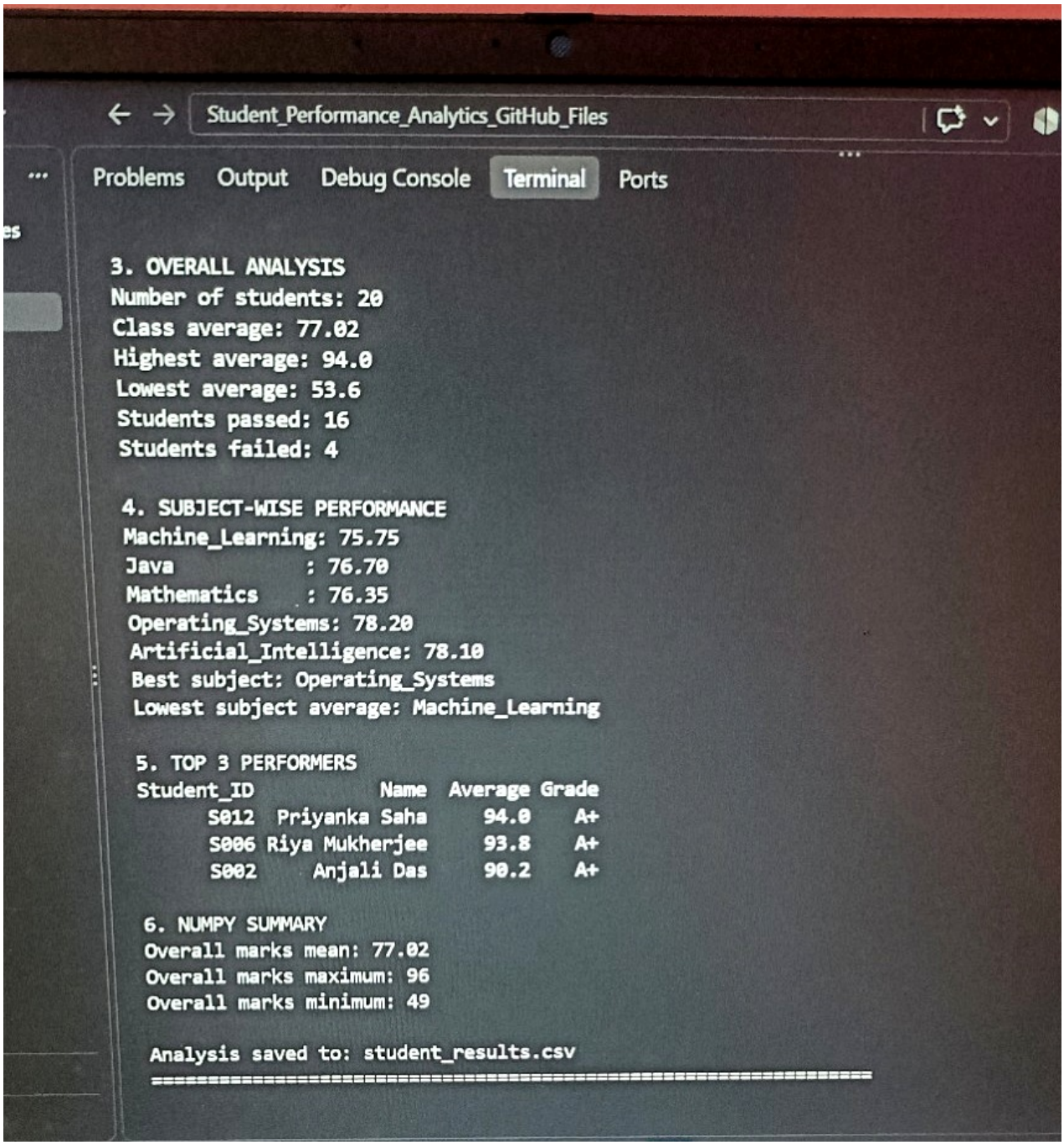
Screenshot 2 — Student Performance Summary

Shows each student's total, average, grade, attendance, and result.



Screenshot 3 — Overall Analysis and Top Performers

Shows class statistics, subject averages, top three performers, NumPy summary, and the saved output file.



8. Conclusion

The project demonstrates the use of Python, Pandas, and NumPy to process student data and create a useful performance summary. It automates routine calculations, compares subject-wise performance, identifies top performers, and exports the analysis to a CSV file.

9. Learning Outcomes

• Working with CSV datasets and Pandas DataFrames.

• Using functions, conditions, and loops in Python.

• Performing numerical calculations with NumPy and Pandas.

• Summarising data and exporting processed results.

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GitHub Repository

The source code and project files for the Student Performance Analytics System are available at the repository below.

https://github.com/sumitsingh-R/Student-Performance-Analytics-System

Repository: Student Performance Analytics System

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